FINITE LOCALLY FREE EQUIVALENCE RELATIONS WITH AN AFFINE-CLASS CONDITION GIVE ALGEBRAIC SPACE QUOTIENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original reference 03JE: sources/stacks-chow/tags.txt, source lines 4636–4636. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02WU: sources/stacks-chow/tags.txt, source lines 3817–3817. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03BM: sources/stacks-chow/tags.txt, source lines 4353–4353. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01JJ: sources/stacks-chow/tags.txt, source lines 2016–2016. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Proposition 0.1. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume*

(1) *$s, t : R \to U$ finite locally free,*
(2) *$j = (t, s)$ is an equivalence relation, and*
(3) *every nonempty closed subset Z of U contains a point u whose R-equivalence class $t(s^{-1}(\{u\}))$ is contained in an affine open of U[1].*

Then there exists a finite locally free morphism $U \to M$ of schemes over S such that $R = U \times_M U$ and such that M represents the quotient sheaf U/R in the fppf topology.

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[1]Let $E \subset U$ be the set of points u such that $t(s^{-1}(\{u\}))$ is contained in an affine open of U. Condition (3) holds if $E = U$, or if every finite type point of U is in E, or if every $u \in U$ specializes to a point of E.

Proof. By assumption (3) and Groupoids, Lemma 03JE we can find an open covering $U = \bigcup U_i$ such that each U_i is an R-invariant affine open of U. Set $R_i = R|_{U_i}$. Consider the fppf sheaves $F = U/R$ and $F_i = U_i/R_i$. By Spaces, Lemma 02WU the morphisms $F_i \to F$ are representable and open immersions. By Groupoids, Proposition 03BM the sheaves F_i are representable by affine schemes. If T is a scheme and $T \to F$ is a morphism, then $V_i = F_i \times_F T$ is open in T and we claim that $T = \bigcup V_i$. Namely, fppf locally on T we can lift $T \to F$ to a morphism $f : T \to U$ and in that case $f^{-1}(U_i) \subset V_i$. Hence we conclude that F is representable by a scheme, see Schemes, Lemma 01JJ. $\square$